ADDING VELOCITIES ON ONE PAGE

Galileo, Lorentz and Euclid

One addition formula with a single sign switch covers rotations, Galilean shears and Lorentz boosts.

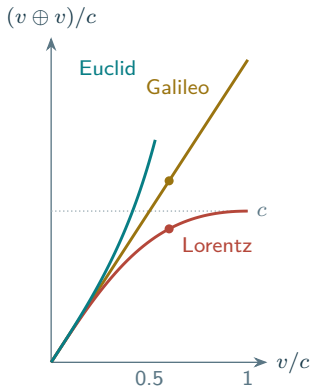
THE RULE

One formula, three geometries

Composing two slopes u and v, whether directions in a plane or velocities in spacetime, follows

$$u \oplus v = \frac{u+v}{1-\kappa uv}, \qquad \kappa = \begin{cases} +1 & \text{Euclidean rotations, } \tan(\alpha+\beta) \\ 0 & \text{Galilean shears, } u+v \\ -1/c^2 & \text{Lorentz boosts, } \dfrac{u+v}{1+uv/c^2} \end{cases}$$

Galileo's rule is the limit $c \to \infty$ of Einstein's, and the two differ only when uv is a noticeable fraction of c^2: two aircraft at $300\,\mathrm{m/s}$ differ from simple addition by one part in 10^{12}, while $0.6c \oplus 0.6c = \frac{15}{17}c \approx 0.882c$ falls well short of $1.2c$.



A slope added to itself: Lorentz never reaches c, while tangents run away.

RAPIDITY

The quantity that does add

Writing $v = c\tanh\varphi$ turns Einstein's rule into the addition formula for $\tanh$, so the rapidities simply add,

$$\tanh(\varphi_1+\varphi_2) = \frac{\tanh\varphi_1 + \tanh\varphi_2}{1+\tanh\varphi_1\tanh\varphi_2}, \qquad \varphi = \operatorname{artanh}\frac{v}{c},$$

just as angles add where tangents do not. For small speeds $\varphi \approx v/c$, which is why everyday velocities seem to add. Rapidity also governs the Doppler factor e^{φ} and the energy and momentum, $E = mc^2\cosh\varphi$ and $p = mc\sinh\varphi$, and a rocket with steady proper acceleration a gains rapidity at the steady rate a/c.

v/c	φ	e^{φ}
0.6	$\ln 2 = 0.693$	2
$0.6 \oplus 0.6 = \frac{15}{17}$	$\ln 4 = 1.386$	4
0.9	1.472	4.36
$0.9 \oplus 0.9 = 0.9945$	2.944	19

Rapidities add and Doppler factors multiply.

EUCLIDEAN, GALILEAN, MINKOWSKIAN

Rotations, shears and boosts as three kinds of complex number

	Euclidean plane	Galilean spacetime	Minkowski spacetime
unit and its square	$\mathrm{i}^2 = -1$	$\varepsilon^2 = 0$	$j^2 = +1$
what is kept	$x^2 + y^2$	time t	$c^2t^2 - x^2$
unit curve	circle	pair of lines $t = \pm 1$	hyperbola
transformation	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ v & 1 \end{pmatrix}$	$\begin{pmatrix} \cosh\varphi & \sinh\varphi \\ \sinh\varphi & \cosh\varphi \end{pmatrix}$
parameter that adds	angle θ	velocity v	rapidity φ
slope it gives	$\tan\theta$	v	$\tanh\varphi = v/c$
exponential form	$\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$	$\mathrm{e}^{\varepsilon v} = 1 + \varepsilon v$	$\mathrm{e}^{j\varphi} = \cosh\varphi + j\sinh\varphi$

Multiplying two such numbers composes the transformations, so the angle, the velocity or the rapidity adds in the exponent; the Galilean column is the degenerate middle case where the second-order term vanishes, the parabolic member of the elliptic, parabolic and hyperbolic trichotomy.

WHY NOTHING OVERTAKES LIGHT

A product that stays positive

In units with $c = 1$, Einstein's rule satisfies the identity

$$1 - w^2 = \frac{(1-u^2)(1-v^2)}{(1+uv)^2}, \qquad w = u \oplus v,$$

so two speeds below light always combine to a speed below light, and $u \oplus 1 = 1$ for every u. The Euclidean twin, $1 + w^2 = \frac{(1+u^2)(1+v^2)}{(1-uv)^2}$, is $\sec^2$ of a sum, which blows up when the angles reach a right angle.

THE FIRST EXPERIMENTAL HINT

Fizeau's moving water, 1851

Light travels through still water at c/n. Fizeau sent it along water flowing at speed v and found it dragged only partly, by the fraction $1 - 1/n^2 \approx 0.44$ that Fresnel had predicted. Expanding Einstein's rule to first order in v gives exactly

$$\frac{c}{n} \oplus v \approx \frac{c}{n} + v\left(1 - \frac{1}{n^2}\right),$$

fifty-four years before special relativity explained it.

Conventions. Velocities lie along one line, ε and j are the dual and split-complex units, and κ selects the geometry.

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